

Jacob W. Toney

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EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Ph.D., Chemical Engineering and Computational Science (Minor: Quantum Chemistry)

Doctoral Advisor: Prof. Heather J. Kulik

Thesis: Accelerating Quantum Chemistry with Machine Learning for the Rational Design of Transition Metal Complexes

Relevant Coursework: Computational Chemistry, Advanced Electronic Structure Theory, Machine Learning Algorithms, ML for Chemistry, Matrix Calculus for ML, Diffusion Models, Generative AI and Stochastic Differential Equations, Numerical Methods, Bayesian Modeling and Inference

University of Southern California (USC)

Los Angeles, CA

B.S., Chemical Engineering (Minor: Materials Science) | GPA: 4.00 Aug. 2017 – May 2021

M.S., Chemical Engineering

Sep. 2019 – May 2021

Research Advisor: Prof. Shaama M. Sharada

Thesis: "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts" (Awarded Best Research Paper)

PUBLICATIONS

1. **J. W. Toney**, A. Farnood, S. Darouich, H. J. Kulik, "Physics-Based Molecular Fingerprints from Spectral Graph Theory", *In Preparation*.
2. A. G. Garrison*, **J. W. Toney***, T. Nikolaeva, R. G. St. Michel, C. J. Stein, H. J. Kulik, "The BOS-TMC Dataset: DFT Properties of 159k Experimentally Characterized Transition Metal Complexes Spanning Multiple Charge and Spin States", *J. Chem. Inf. Model.* (Accepted), **2026**, 10.48550/arXiv.2604.07623.
3. S. Darouich, **J. W. Toney**, W. Luo, J. Kästner, M. Niepert, H. J. Kulik, "Beyond the Training Domain: Robust Generative Transition State Models for Unseen Chemistry", *Nat. Comput. Sci.* (Accepted), **2026**, 10.48550/arXiv.2601.16469.
4. **J. W. Toney***, S. Darouich*, Y. Wang, A. G. Garrison, J. Kästner, H. J. Kulik, "ElemeNet: Multiscale Molecular Machine Learning with Uncertainty Quantification Across the Periodic Table", *arXiv*, **2026**, 10.48550/arXiv.2606.30961.
5. G. G. Terrones, R. G. St. Michel, **J. W. Toney**, A. K. Ball, Y. Wang, A. G. Garrison, A. Nandy, R. Meyer, F. Edholm, C. Oh, S. G. Pujet, D. B. K. Chu, D. Muhammetgulyyev, H. J. Kulik, "molSimplify 2.0: Improved Structure Generation for Automating Discovery in Inorganic Molecular and Reticular Chemistry", *J. Chem. Inf. Model.* **2026**, 10.1021/acs.jcim.5c02733.
6. Y. Liu, **J. W. Toney**, J. M. Cavanagh, K. Sun, A. L. Smith, C. Yuan, R. G. St. Michel, P. A. Graggs, D. Toste, H. J. Kulik, T. Head-Gordon, "Exploring Transition Metal Complexes with Large Language Models", *ChemRxiv* **2025**, 10.26434/chemrxiv-2025-hm3zb.
7. **J. W. Toney**, R. G. St. Michel, A. G. Garrison, I. Kevlishvili, H. J. Kulik, "Identifying Dynamic Metal–Ligand Coordination Modes with Ensemble Learning", *J. Am. Chem. Soc.* **2025**, 10.1021/jacs.5c17169.

8. **J. W. Toney**, R. G. St. Michel, A. G. Garrison, I. Kevlishvili, H. J. Kulik, "Graph neural networks for predicting metal–ligand coordination of transition metal complexes", *Proc. Natl. Acad. Sci.* **2025**, 122, 41, 10.1073/pnas.2415658122.
9. **J. W. Toney**, A. G. Garrison, W. Luo, R. G. St. Michel, S. Mukhopadhyay, and H. J. Kulik, "Exploring beyond experiment: generating high-quality datasets of transition metal complexes with quantum chemistry and machine learning", *Curr. Opin. Chem. Eng.* **2025**, 50, 10.1016/j.coche.2025.101189.
10. I. Kevlishvili, R. G. St. Michel, A. G. Garrison, **J. W. Toney**, H. Adamji, H.-J. Jia, Y. Román-Leshkov, H. J. Kulik, "Leveraging natural language processing to curate the tmCAT, tmPHOTO, tmBIO, and tmSCO datasets of functional transition metal complexes", *Faraday Discuss.* **2025**, 256, 275-303, 10.1039/D4FD00087K.
11. Z. Lan*, **J. W. Toney***, S. M. Sharada, "A computational mechanistic study of CH hydroxylation with mononuclear copper-oxygen complexes", *Catal. Sci. Technol.* **2023**, 13, 342-351, 10.1039/D2CY01128J

HONORS & AWARDS

- 2026** Entalpic AI for Science Fellowship, Entalpic, April 2026
- 2025** Leslye Miller Fraser and Darryl M. Fraser Fellow, MIT School of Engineering, May 2025
- 2024** MathWorks Engineering Fellow, MIT Department of Chemical Engineering, June 2024
- 2022** Alfred P. Sloan UCEM Fellowship, MIT School of Engineering, September 2022
- 2021** Albert Dorman Valedictorian, USC Viterbi School of Engineering, May 2021
 National Academy of Engineering Grand Challenges Scholar, USC Viterbi School of Engineering, May 2021
 University Trustees Award (awarded for perfect GPA), USC, May 2021
 Best Research Paper, "Bioinspired Catalysis of C–H Hydroxylation: A Computational Approach", USC Viterbi School of Engineering, May 2021
 Best Analytical & Expository Essay, "Engineering Ethics: Grand Challenges and Grand Responsibilities", USC Viterbi School of Engineering, May 2021
 Outstanding Chemical Engineering Student Award, USC Department of Chemical Engineering and Materials Science, May 2021
 Best Research Presentation, "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts", USC Department of Chemical Engineering and Materials Science, February 2021
- 2019** Provost's Undergraduate Research Fellowship, USC, January 2019 – December 2020
- 2017** Presidential Scholar, USC, Aug. 2017
 Latino Alumni Association Scholar, USC, Aug. 2017

PRESENTATIONS

1. International Conference on Quantum Chemistry, Berkeley, CA. "Machine Learning Prediction of Structural and Electronic Properties of Transition Metal Complexes", poster. June 2026.

2. LeMaterial Reading Group, Virtual. "Exploring Structure, Properties, and Reactivity in Transition Metal Complexes with Deep Learning and Quantum Chemistry", online talk. February 2026.
3. American Institute of Chemical Engineers (AIChE), Boston, MA. "Graph neural networks for predicting metal–ligand coordination in transition metal complexes", talk in *Data Science and ML Approaches to Catalysis: Catalyst Design and Discovery*. November 2025.
4. World Association of Theoretical and Computational Chemists (WATOC), Oslo, Norway. "Graph neural networks for predicting metal–ligand coordination in transition metal complexes", talk in *Chemical Structure and Reactivity: Analysis*. June 2025.
5. CECAM Frontiers of Computational Reaction Prediction, Chicago, IL. "Graph neural networks for predicting metal–ligand coordination in transition metal complexes", poster. July 2024.
6. USC Department of Chemical Engineering and Materials Science Research Symposium, Los Angeles, CA (virtual). "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts", recorded talk. February 2021. **Awarded Best Research Presentation.**
7. AIChE Annual Student Conference, San Francisco, CA (virtual). "Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes", poster. November 2020.
8. USC Department of Chemical Engineering and Materials Science Research Symposium, Los Angeles, CA. "Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes", poster. March 2020.
9. AIChE Annual Student Conference, Orlando, FL. "Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes", poster. November 2019.

PROFESSIONAL EXPERIENCE

IBM Research Algorithms and Applications Intern (Internship)	Yorktown Heights, NY Jun. 2026 – Aug. 2026
Aspen Technology Software Engineering Intern (MIT Practice School)	Bedford, MA Jul. 2023 – Aug. 2023
Woodside Energy Engineering & Data Science Intern (MIT Practice School)	Perth, WA, Australia Jun. 2023 – Jul. 2023
SpaceX Operations Engineer (Full-time)	Hawthorne, CA Jun. 2021 – Aug. 2022
ExxonMobil Research & Technology Development Process Engineering R&D Intern (Internship)	Houston, TX Jun. 2020 – Aug. 2020
California Resources Corporation Facilities Engineering Intern: Gas Plants (Internship)	Bakersfield, CA May 2019 – Aug. 2019

TEACHING & COURSEWORK

5.698/10.637: Computational Chemistry (MIT) Grader. Graduate elective in chemical engineering and chemistry.	Cambridge, MA Sep. 2025 – Dec. 2025
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| 10.34: Numerical Methods Applied to Chemical Engineering (MIT) | Cambridge, MA |
| Graduate Teaching Fellow. Graduate chemical engineering core course. | Sep. 2024 – Dec. 2024 |
| Instructor rating: 6.5/7 | |
| 10.65: Chemical Reactor Engineering (MIT) | Cambridge, MA |
| Grader. Graduate chemical engineering core course. | Feb. 2024 – May 2024 |
| 10.40: Chemical Engineering Thermodynamics (MIT) | Cambridge, MA |
| Grader. Graduate chemical engineering core course. | Sep. 2023 – Dec. 2023 |

MENTORSHIP & SERVICE

Undergraduate Students

- David Sauer (Chemistry, Harvard University)
Project: Reparameterization of electronic structure methods for transition metal complexes.
- Zheming Zhang (Chemical Engineering, Massachusetts Institute of Technology)
Project: Benchmarking deep learning architectures for electronic property prediction in variable charge and spin states.
- Lauren Wright (Chemical Engineering, Louisiana State University)
Project: Designing new hydroformylation catalysts through mechanistic study and data science.
Presentation: AIChE Annual Student Conference, Boston, MA. "Leveraging Computational Chemistry and Data Science to Design Novel Hydroformylation Catalysts", poster. November 2025. **Awarded Third Place Poster.**
- Sebastian Pujet (Chemical Engineering and Computer Science, Washington University in Saint Louis)
Project: Active learning to optimize mechanochemical properties of transition metal complexes.
Contributed Publications: G. G. Terrones, R. G. St. Michel, J. W. Toney, A. K. Ball, Y. Wang, A. G. Garrison, A. Nandy, R. Meyer, F. Edholm, C. Oh, S. G. Pujet, D. B. K. Chu, D. Muhammetgulyyev, H. J. Kulik, "molSimplify 2.0: Improved Structure Generation for Automating Discovery in Inorganic Molecular and Reticular Chemistry", *J. Chem. Inf. Model.* **2026**, 10.1021/acs.jcim.5c02733.

USC AIChE Student Chapter

President

Los Angeles, CA
Apr. 2020 – May 2021

USC Latino Alumni Association Scholar Committee

President

Los Angeles, CA
Jan. 2019 – May 2021